ID3802 – Monthly Progress Report (February)

1 Work done in the current month

1.1 Literature Survey

We read the work of Peters on Online Elicitation of Optimal Matchings [1].Our original algorithm is essentially inspired from the open question proposed by Peters in his paper on Online Elicitation of Necessarily Optimal Matchings. It is very fundamental to comprehend his algorithms, the corresponding optimal offline algorithms and the techniques he employed to come out with competitive ratios(α) for moving forward. He gives algorithms for online elicitation of necessarily rank-maximal matchings based on both next-best query model and hybrid query model. Our primary focus was on the online algorithm he proposed based on the next-best query model, the proof of correctness, and competitive ratio analysis for the same.

The algorithm is followed by the correctness proof of the algorithm. The author introduces a couple of terms and related constraints which are helpful in the upcoming α - analysis. After setting up the plot, the author goes for deriving a upper bound of 3 for α. This is followed by the rigorous analysis for the tight upper bound of 3/2.

1.2 Plot

Lemma 1.1. *Let $rev(A)$ denotes the set of agents who are matched to a revealed post. Then in any instance of necessarily rank-maximal matching problem, every $a \in rev(A)$ must have at least $r_a - 1$ revealed posts, where r_a is the first iteration where a becomes an odd or unreachable agent for the first time.*

That is, atleast $|r_a - 1|$ queries are needed for each agent a by the optimal algorithm to determine the NRM.

1.3 Upper bound

The author adopts a classic strategy in competitive analysis. This approach starts by kind of glorifying the OPT to create a safety margin (constant-competitive) and later applies the real structural constraints and thus grounds the adversary.
It starts with some assumptions for this loose analysis

- The optimal algorithm should match at least $n - 1$ agents to revealed posts
- If an agent is matched to a revealed post, $r_a - 1$ preferences must be revealed for this particular agent
- In our algorithm, each agent has to reveal r_a posts

So,

$$ALG \leq \Sigma(r_a) + n - 1$$

and

$$OPT \geq \Sigma_{a \in A'}(max(r_a - 1, 1))$$

Thus we get

$$\frac{ALG}{OPT} \leq \frac{\sum_{a \in A'}(r_a) + n - 1}{\sum_{a \in A'}(\max(r_a - 1, 1))} \leq \frac{\sum_{a \in A'}(r_a + 1)}{\sum_{a \in A'}(\max(r_a - 1, 1))}$$

$$\frac{\sum_{a \in A'}(r_a + 1)}{\sum_{a \in A'}(\max(r_a - 1, 1))} \leq \max_{a \in A'}\left(\frac{r_a + 1}{\max(r_a - 1, 1)}\right) \leq 3$$

1.4 Tight analysis for 3/2

Here we employ more rigorous structural comparisons and derive the tight bound of 3/2 for α. Instead of isolating agents, the method accounts for **resource conflicts** by **grouping agents** who compete for the same house which forces the OPT also to work harder to resolve conflicts.

1.4.1 The unrevealed case

Suppose agent a' matches to an unrevealed house. Then this leftover house should be the last preference for all other agents or none of the houses in the revealed matchings are revealed by a'. Otherwise they could swap houses to make a better matching.

Here, OPT has to query every preference of each agent who match to a revealed house to make sure that the leftover house is their worst preference and ALG seeks every preference of all agents.
OPT $= (n-1)^2$
ALG $= n(n-1)$

$$\frac{ALG}{OPT} = \frac{n}{n-1}$$

which is at most 3/2 for $n \geq 2$

1.4.2 Grouping by first choice

The author groups agents by the first-choice house to capture the resource contention. A_h is defined as the set of all agents whose first choice is house h. OPT should ask at least $2|A_h|$-1 queries to this group which can be interpreted as two queries each to all the losers who have better fallback option as the second choice and one query to the winner. $OPT(A_h)$ is the minimum number of queries the optimal algorithm must ask the specific group of agents who rank house h as their first choice.

$$OPT(A_h) \geq max(\Sigma_{a \in A_h}(max(r_a - 1, 1), 2|A_h| - 1)$$
$$= \max\left(\text{individual necessity}, \text{group conflict}\right)$$
$$OPT = \Sigma_{h \in H} OPT(A_h)$$
$$ALG = \Sigma_{h \in H} \Sigma_{a \in A} r_a$$

The author applies the powerful **mediant rule** ie.,

$$\frac{\Sigma a_i}{\Sigma b_i} \leq max(\frac{a_i}{b_i})$$

Thus the upper bound reduces to

$$\frac{OPT}{ALG} \leq max_{h \in H} \frac{\Sigma_{a \in A_h} r_a}{OPT(A_h)}$$

Now we have two cases :

1. **High query case:** $\sum_{a \in A_h} r_a \geq 3|A_h|$

$$\frac{OPT}{ALG} \leq \max_{h \in H} \frac{\sum_{a \in A_h} r_a}{OPT(A_h)} \leq \max_{h \in H} \frac{\sum_{a \in A_h} r_a}{\max\left(\sum_{a \in A_h} \max(r_a - 1, 1),\ 2|A_h| - 1\right)} \leq \max_{h \in H} \frac{\sum_{a \in A_h} r_a}{\sum_{a \in A_h} \max(r_a - 1, 1)}$$

$$\frac{OPT}{ALG} \leq \max_{h \in H} \frac{\sum_{a \in A_h} r_a}{r_a - |A_h|} \leq \frac{\sum_{a \in A_h} r_a}{r_a - \frac{1}{3}\sum_{a \in A_h} r_a} = \frac{\sum_{a \in A_h} r_a}{\frac{2}{3}\sum_{a \in A_h} r_a} = \frac{3}{2}.$$

2. **Low query case:** $\sum_{a \in A_h} r_a < 3|A_h|$
   There exists at least one $a \in A_h$ with $r_a = 2$ since $\sum_{a \in A_h} r_a < 3|A_h|$.
   Let $A_h^2 = \{a \in A_h \mid r_a = 2\}$.
   (a) $|A_h^2| \geq 2$

$$\frac{\sum_{a \in A_h} r_a}{\mathrm{OPT}(A_h)} = \frac{\sum_{a \in A_h \setminus A_h^2} r_a + 2|A_h^2|}{\mathrm{OPT}(A_h \setminus A_h^2) + \mathrm{OPT}(A_h^2)}$$
$$\leq \max\left(\frac{\sum_{a \in A_h \setminus A_h^2} r_a}{\sum_{a \in A_h \setminus A_h^2} r_a - |A_h \setminus A_h^2|}, \frac{2|A_h^2|}{2|A_h^2| - 1}\right) \leq \frac{3}{2}.$$

   Since $\sum_{a \in A_h \setminus A_h^2} r_a \geq 3|A_h \setminus A_h^2|$ and it had already been proven that for such a case, $\frac{ALG}{OPT} \leq \frac{3}{2}$, the first argument for max is bounded by $\frac{3}{2}$.

$$\frac{2|A_h^2|}{2|A_h^2| - 1} \leq \frac{3}{2} \quad \text{for all } A_h \text{ such that } |A_h| \geq 2.$$

   Therefore, we get an upper bound of $\frac{3}{2}$.

(b) $|A_h^2| = 1$

There is exactly one $a \in A_h$ with $r_a = 2$. So a is O or U in the 2nd iteration. Since all other agents are E in the 2nd iteration and they all have edges to h, h can't be even in the second iteration since EE edges are not allowed. So (a, h) Since h is unavailable, and a gets out of E in the 2nd iteration, a must be matched to it's 2nd preference. So, OPT must ask a 2 queries.

$$\frac{ALG}{OPT} \leq \frac{\sum_{a \in A_h} r_a}{\max\left(\sum_{a \in A_h} \max(r_a - 1, 1),\ 2|A_h| - 1\right)} \leq \frac{3|A_h| - 1}{2|A_h| - 1} \approx \frac{3|A_h| - 1}{2|A_h|} = \frac{3 - \frac{1}{|A_h|}}{2} \leq \frac{3}{2}.$$

Thus, the algorithm has a competitive ratio of $\frac{3}{2}$.

2 Feedback from the mentor

Satisfactory.

3 Signature of the mentor

Pratik Ghosal

01/03/2026

References

[1] Jannik Peters. Online elicitation of necessarily optimal matchings. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 36, pages 5164–5172, 2022.